

```
In[24]:= x = .; Remove["Global`*"]; Off[General::spell1]; Off[General::spell];

In[25]:= conjugateRule={Complex[re_,im_]>Complex[re,-im]};
conjugate[z_]:=z/.conjugateRule;
real[z_]:= (1/2)*(z+conjugate[z])//Simplify;
imag[z_]:= (-I/2)*(z-conjugate[z])//Simplify;
abs[z_]:=Sqrt[z*(z/.conjugateRule)];
```

Constants

```
In[30]:= c = 299 792 458 ;
hb = 1.0545715964207855 * 10^-34;
q = 1.602 * 10^-19;
e0 = 8.85 * 10^-12;
m = 9.1 * 10^-31;
a0 = 3.5668 * 10^-10;
den = 8 / a0^3;
G = Sqrt[1^2 + 1^2 + 3^2] * 2 * Pi / a0;
normstructurefactor = 8 / 8;
atomscatfactor = 1.603;
re = 2.82 * 10^-13 * 10^-2;
e0 = 8.85 * 10^-12;
m = 9.1 * 10^-31;
```

Scattering Factors (Carbon)

Note : Carbon data are hidden in the following cell.

```
In[32]:= listf1 = carbondata[[All, {2, 3}]]
listf2 = carbondata[[All, {2, 4}]]
f1[w_] := Interpolation[listf1, InterpolationOrder -> 4][w]
f2[w_] := Interpolation[listf2, InterpolationOrder -> 4][w]

Out[32]=
```

$$\left\{ \left\{ 4.451 \times 10^{16}, 3.6332 \right\}, \left\{ 4.523 \times 10^{16}, 3.6738 \right\}, \left\{ 4.5961 \times 10^{16}, 3.7127 \right\}, \right. \\ \left. \left\{ 4.6705 \times 10^{16}, 3.7475 \right\}, \left\{ 4.746 \times 10^{16}, 3.7797 \right\}, \left\{ 4.8228 \times 10^{16}, 3.8097 \right\}, \right. \\ \left. \left\{ 4.9008 \times 10^{16}, 3.8376 \right\}, \left\{ 4.98 \times 10^{16}, 3.8637 \right\}, \left\{ 5.0606 \times 10^{16}, 3.8881 \right\}, \right. \\ \left. \left\{ 5.1424 \times 10^{16}, 3.9107 \right\}, \left\{ 5.2256 \times 10^{16}, 3.9316 \right\}, \left\{ 5.3101 \times 10^{16}, 3.9503 \right\}, \right. \\ \left. \left\{ 5.396 \times 10^{16}, 3.9671 \right\}, \left\{ 5.4833 \times 10^{16}, 3.9849 \right\}, \left\{ 5.572 \times 10^{16}, 4.0025 \right\}, \right. \\ \left. \left\{ 5.6621 \times 10^{16}, 4.0199 \right\}, \left\{ 5.7537 \times 10^{16}, 4.0371 \right\}, \left\{ 5.8467 \times 10^{16}, 4.0543 \right\}, \right. \\ \left. \left\{ 5.9413 \times 10^{16}, 4.0736 \right\}, \left\{ 6.0374 \times 10^{16}, 4.0945 \right\}, \left\{ 6.1351 \times 10^{16}, 4.1101 \right\}, \right. \\ \left. \left\{ 6.2343 \times 10^{16}, 4.1223 \right\}, \left\{ 6.3351 \times 10^{16}, 4.1321 \right\}, \left\{ 6.4376 \times 10^{16}, 4.1399 \right\}, \right. \\ \left. \left\{ 6.5417 \times 10^{16}, 4.1456 \right\}, \left\{ 6.6475 \times 10^{16}, 4.1493 \right\}, \left\{ 6.755 \times 10^{16}, 4.1495 \right\}, \right. \\ \left. \left\{ 6.8643 \times 10^{16}, 4.1458 \right\}, \left\{ 6.9753 \times 10^{16}, 4.1417 \right\}, \left\{ 7.0881 \times 10^{16}, 4.1417 \right\}, \right. \\ \left. \left\{ 7.2028 \times 10^{16}, 4.143 \right\}, \left\{ 7.3193 \times 10^{16}, 4.1449 \right\}, \left\{ 7.4377 \times 10^{16}, 4.147 \right\}, \right. \\ \left. \left\{ 7.5579 \times 10^{16}, 4.1491 \right\}, \left\{ 7.6802 \times 10^{16}, 4.1511 \right\}, \left\{ 7.8044 \times 10^{16}, 4.1528 \right\}, \right. \\ \left. \left\{ 7.9306 \times 10^{16}, 4.1537 \right\}, \left\{ 8.0589 \times 10^{16}, 4.1531 \right\}, \left\{ 8.1893 \times 10^{16}, 4.1496 \right\}, \right\}$$

$\{8.3217 \times 10^{16}, 4.15\}$, $\{8.4563 \times 10^{16}, 4.1527\}$, $\{8.5931 \times 10^{16}, 4.1568\}$,
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$\{3.4694 \times 10^{19}, 6.0013\}$, $\{3.5255 \times 10^{19}, 6.0012\}$, $\{3.5825 \times 10^{19}, 6.0012\}$,
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 $\{4.4134 \times 10^{19}, 6.0001\}$, $\{4.4848 \times 10^{19}, 6.0001\}$, $\{4.5573 \times 10^{19}, 6.\}$

Out[33]=

$\{4.451 \times 10^{16}, 2.7402\}$, $\{4.523 \times 10^{16}, 2.6898\}$, $\{4.5961 \times 10^{16}, 2.6362\}$,
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```

```
In[36]:=
```

Refractive index and loss

```
In[37]:= n[w_] := 1 - den * re * 2 * Pi * c^2 / w^2 * f1[w];
          α[w_] := w / c * den * re * 2 * Pi * c^2 / w^2 * f2[w];
```

Parameters

```
In[39]:= pumpphotonenergy = 9.817 * 10^3;
          wLASER = pumpphotonenergy * q / hb;
          wSHG = 2 * wLASER;
          kLASER = wLASER / c * n[wLASER]; kSHG = wSHG / c * n[wSHG];
          L = 0.8 * 10^-3;
```

```
In[44]:= energy = 50 * 10^-6;
```



```
In[45]:= photonsperpulse = energy / (hb * wLASER)
```

```
Out[45]=  
 $3.17928 \times 10^{10}$ 
```

```
In[46]:= w0 = 5 * 10^-6;  
tau = 3 * 10^-14;  
bandwidth = 0.5 * 10^-4 * wLASER;  
theta = thetaSHG0;  
thetaP = thetaLASER0;  
(*vg=veff;*)  
kappa = kappa;  
alphaw = alpha[wLASER];  
deltaKx = deltaKx1[0, deltaTheta];  
deltaKz = deltaKz1[0, deltaTheta];
```

Find Base values

```
In[54]:= thetaB = Abs[ArcSin[G / (2 * kSHG)]];]
```

```
thetaB * 180 / Pi
```

```
Out[55]=  
17.0751
```

```
In[56]:= eq1 = 2 * kLASER * Sin[ (thetaLASER) ] + G - kSHG * Sin[thetaSHG]
```

```
Out[56]=  
 $5.84248 \times 10^{10} + 9.94881 \times 10^{10} \sin[\theta_{\text{LASER}}] - 9.94887 \times 10^{10} \sin[\theta_{\text{SHG}}]$ 
```

```
In[57]:= eq2 = 2 * kLASER * Cos[ (thetaLASER) ] - kSHG * Cos[thetaSHG]
```

```
Out[57]=  
 $9.94881 \times 10^{10} \cos[\theta_{\text{LASER}}] - 9.94887 \times 10^{10} \cos[\theta_{\text{SHG}}]$ 
```

```
In[58]:= sol = Solve[{eq1 == 0, eq2 == 0}, {thetaLASER, thetaSHG}]
```

 **Solve:** Inverse functions are being used by Solve, so some solutions may not be found; use Reduce for complete solution information. 

```
Out[58]=  
{ {thetaLASER -> -2.84358, thetaSHG -> 2.84357}, {thetaLASER -> -0.298009, thetaSHG -> 0.298027} }
```

```
In[59]:= thetaLASER0 = thetaLASER /. sol[[2, 1]]
```

```
Out[59]=  
-0.298009
```

```
In[60]:= thetaSHG0 = thetaSHG /. sol[[2, 2]]
```

```
Out[60]=  
0.298027
```

Kappa

```
In[61]:= kappa = 1;
```

```
In[62]:= kappa = kappa;
```

Delta K

```
In[63]:= eq11 =
          2 * (wLASER) * n[wLASER] * Sin[θLASER0 + δθ] / c + G - wSHG * n[wSHG] * Sin[θSHG0 + δ2] / c;

In[64]:= sol = Solve[eq11 == 0, δ2]

 $\dots$  Solve: Inverse functions are being used by Solve, so some solutions may not be found; use Reduce for complete solution
information. ⓘ

Out[64]=
{{δ2 → -0.298027 + 1. ArcSin[0.587251 - 0.999994 Sin[0.298009 - 1. δθ]]}}
```

```
In[65]:= δ22[δθ_] = δ2 /. sol[[1, 1]]
Out[65]=
-0.298027 + 1. ArcSin[0.587251 - 0.999994 Sin[0.298009 - 1. δθ]]

In[66]:= δkz1[δw_, δθ_] = 2 * (wLASER + δw) * n[wLASER + δw] * Cos[θLASER0 + δθ] / c -
          wSHG * n[wSHG] * Cos[θSHG0 + δ22[δθ]] / c;

In[67]:= δkx1[δw_, δθ_] = 2 * (wLASER + δw) * n[wLASER + δw] * Sin[θLASER0 + δθ] / c +
          G - wSHG * n[wSHG] * Sin[θSHG0 + δθ] / c;
```

Group velocity

```
In[68]:= k1[w_] := w * n[w] / c
In[69]:= vg[w_] := 1 / D[k1[w], w]

In[70]:= vgLASER = 1 / D[k1[w], w] /. w → wLASER
          vgSHG = 1 / D[k1[w], w] /. w → wSHG
Out[70]=
2.9979 × 108
Out[71]=
2.99792 × 108

In[72]:= veff = 1 / (1 / vgSHG - Cos[θ - θp] / vgLASER)
Out[72]=
1.73865 × 109
```

Group velocity

```
In[73]:= k1[w_] := w * n[w] / c
In[74]:= vg[w_] := 1 / D[k1[w], w]
```

In[75]:= **vgLASER = 1 / D[k1[w], w] /. w → wLASER**

vgSHG = 1 / D[k1[w], w] /. w → wSHG

Out[75]=

$$2.9979 \times 10^8$$

Out[76]=

$$2.99792 \times 10^8$$

In[77]:= **veff = 1 / (1 / vgSHG - Cos[θ - θp] / vgLASER)**

Out[77]=

$$1.73865 \times 10^9$$

SHG efficiency (From SHG FOURIER METHOD)

In[78]:= **ξ = 0;**

alphaw = 0;

In[11]:= **Simplify** $\left[4 e^{\frac{(2 k x^2 w \theta^2 - 2 i k x^2 w \theta^2 \xi + w^2 \tau^2 \cos[2 \theta p]) \sec[\theta p]^2}{16 (-1 + i \xi)} + \frac{(2 k x^2 w \theta^2 + 2 i k x^2 w \theta^2 \xi + w^2 \tau^2 \cos[2 \theta p]) \sec[\theta p]^2}{16 (-1 - i \xi)}} \right.$

$$\left. \left(-1 + e^{\frac{z \sec[\theta] \sec[\theta p] (-2 \text{alphaw} \text{veff} \cos[\theta] + i \text{veff} \delta k z \cos[\theta - \theta p] - 2 i w \cos[\theta p] + i \text{veff} \delta k z \cos[\theta + \theta p] - 2 i k x \text{veff} \sin[\theta - \theta p])}{2 \text{veff}}} \right) \right.$$

$$\left. \left(-1 + e^{\frac{z \sec[\theta] \sec[\theta p] (-2 \text{alphaw} \text{veff} \cos[\theta] - i \text{veff} \delta k z \cos[\theta - \theta p] + 2 i w \cos[\theta p] - i \text{veff} \delta k z \cos[\theta + \theta p] + 2 i k x \text{veff} \sin[\theta - \theta p])}{2 \text{veff}}} \right) \right.$$

$$\left. \text{kapa}^2 \text{veff}^2 w \theta^2 \cos[\theta] \cos[\theta p]^2 \right) /$$

$$\left(\pi^3 \sqrt{\frac{1 - i \xi}{\tau^2}} \sqrt{\frac{1 + i \xi}{\tau^2}} \text{Abs}[w \theta^2 \tau \cos[\theta p]]^2 (1 + \cos[2 \theta p]) (-2 i \text{alphaw} \text{veff} \cos[\theta] + \right.$$

$$\text{veff} \delta k z \cos[\theta - \theta p] - 2 w \cos[\theta p] + \text{veff} \delta k z \cos[\theta + \theta p] - 2 k x \text{veff} \sin[\theta - \theta p])$$

$$(2 i \text{alphaw} \text{veff} \cos[\theta] + \text{vg} \delta k z \cos[\theta - \theta p] - 2 w \cos[\theta p] +$$

$$\left. \left. \text{veff} \delta k z \cos[\theta + \theta p] - 2 k x \text{veff} \sin[\theta - \theta p]) \right) \right]$$

Out[11]=

$$\left(e^{\frac{1}{8} \left(-8 i z \delta k z - w^2 \tau^2 + \frac{8 i w z \sec[\theta]}{\text{veff}} - (2 k x^2 w \theta^2 + w^2 \tau^2) \sec[\theta p]^2 + 8 i k x z \tan[\theta] - 8 i k x z \tan[\theta p] + w^2 \tau^2 \tan[\theta p]^2 \right)} \right.$$

$$\left. \left(-1 + e^{\frac{i z (-w \sec[\theta] + \text{veff} (\delta k z - k x \tan[\theta] + k x \tan[\theta p]))}{\text{veff}}} \right)^2 \text{kapa}^2 \text{veff}^2 w \theta^2 \tau^2 \sec[\theta p] \right) / \left(\pi^3 \text{Abs}[w \theta^2 \tau \cos[\theta p]]^2 \right.$$

$$(\text{vg} \delta k z \cos[\theta - \theta p] - 2 w \cos[\theta p] + \text{veff} \delta k z \cos[\theta + \theta p] - 2 k x \text{veff} \sin[\theta - \theta p])$$

$$\left. (w \sec[\theta] - \text{veff} (\delta k z - k x \tan[\theta] + k x \tan[\theta p])) \right)$$

$$\begin{aligned}
In[*] := \text{power}[z_, kx_, w_] = & \left(4 e^{\frac{(2 kx^2 w \theta^2 - 2 i kx^2 w \theta^2 \xi + w^2 \tau^2 + w^2 \tau^2 \cos[2 \theta p]) \sec[\theta p]^2}{16 (-1 + i \xi)} + \frac{(2 kx^2 w \theta^2 + 2 i kx^2 w \theta^2 \xi + w^2 \tau^2 + w^2 \tau^2 \cos[2 \theta p]) \sec[\theta p]^2}{16 (-1 - i \xi)}} \right. \\
& \left(-1 + e^{\frac{z \sec[\theta] \sec[\theta p] (-2 \text{alphaw} \text{veff} \cos[\theta] + i \text{veff} \delta k z \cos[\theta - \theta p] - 2 i w \cos[\theta p] + i \text{veff} \delta k z \cos[\theta + \theta p] - 2 i kx \text{veff} \sin[\theta - \theta p])}{2 \text{veff}}} \right) \\
& \left(-1 + e^{\frac{z \sec[\theta] \sec[\theta p] (-2 \text{alphaw} \text{veff} \cos[\theta] - i \text{veff} \delta k z \cos[\theta - \theta p] + 2 i w \cos[\theta p] - i \text{veff} \delta k z \cos[\theta + \theta p] + 2 i kx \text{veff} \sin[\theta - \theta p])}{2 \text{veff}}} \right) \text{kapa}^2 \\
& \text{veff}^2 w \theta^2 \cos[\theta] \cos[\theta p]^2 \Bigg/ \left(\pi^3 \sqrt{\frac{1 - i \xi}{\tau^2}} \sqrt{\frac{1 + i \xi}{\tau^2}} \text{Abs}[w \theta^2 \tau \cos[\theta p]]^2 \right. \\
& (1 + \cos[2 \theta p]) (-2 i \text{alphaw} \text{veff} \cos[\theta] + \text{veff} \delta k z \cos[\theta - \theta p] - 2 w \cos[\theta p] + \\
& \text{veff} \delta k z \cos[\theta + \theta p] - 2 kx \text{veff} \sin[\theta - \theta p]) (2 i \text{alphaw} \text{veff} \cos[\theta] + \\
& \left. \text{vg} \delta k z \cos[\theta - \theta p] - 2 w \cos[\theta p] + \text{veff} \delta k z \cos[\theta + \theta p] - 2 kx \text{veff} \sin[\theta - \theta p]) \right);
\end{aligned}$$